

DGTN: Graph-Enhanced Transformer with Diffusive Attention Gating Mechanism for Enzyme $\Delta\Delta G$ Prediction

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Abstract

Predicting the effect of amino acid mutations on enzyme thermodynamic stability ($\Delta\Delta G$) is fundamental to protein engineering and drug design. While recent deep learning approaches have shown promise, they often process sequence and structure information independently, failing to capture the intricate coupling between local structural geometry and global sequential patterns. We present **DGTN** (Diffused Graph-Transformer Network), a novel architecture that co-learns graph neural network (GNN) weights for structural priors and transformer attention through a diffusion mechanism. Our key innovation is a *bidirectional diffusion process* where: (1) GNN-derived structural embeddings guide transformer attention via learnable diffusion kernels, and (2) transformer representations refine GNN message passing through attention-modulated graph updates. We provide rigorous mathematical analysis showing this co-learning scheme achieves provably better approximation bounds than independent processing. On ProTherm and SKEMPI benchmarks, DGTN achieves state-of-the-art performance (Pearson $\rho = 0.87$, RMSE = 1.21 kcal/mol), with 6.2% improvement over best baselines. Ablation studies confirm the diffusion mechanism contributes 4.8 points to correlation. Our theoretical analysis proves the diffused attention converges to optimal structure-sequence coupling, with convergence rate $O(1/\sqrt{T})$ where T is diffusion steps. This work establishes a principled framework for integrating heterogeneous protein representations through learnable diffusion.

1 Introduction

Motivation and Background

The Gibbs free energy change upon mutation ($\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$) governs protein thermodynamic stability, a critical determinant of enzyme function, disease pathogenesis, and therapeutic efficacy Tokuriki and Tawfik [2009]. Accurate $\Delta\Delta G$ prediction enables rational protein design for industrial biocatalysis Braun et al. [2024, 2025], Listov et al. [2025], therapeutic antibody engineering Jain et al. [2017], and understanding disease mechanisms Yue et al. [2005], Bashour et al. [2024].

Traditional computational approaches employ physics-based energy functions (FoldX [Schymkowitz et al., 2005], Rosetta [Kellogg et al., 2011]), achieving limited accuracy (Pearson $\rho \approx 0.5$) due to approximations in energy calculations and conformational sampling. Recent machine learning methods leverage either sequence information via protein language models [Meier et al., 2021] or structural data through 3D convolutional networks [Zhou et al., 2023], but typically process these modalities separately.

Key Challenges

There are three key challenges when deploying a transformer-based learning algorithm to perform highly accurate $\Delta\Delta$ predictions. **1) Modal Heterogeneity.** Sequence data (1D) and structural data (3D graph)

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have fundamentally different mathematical representations. Naive concatenation or late fusion fails to capture cross-modal dependencies. **2) Local-Global Coupling.** Mutation effects involve both local geometric perturbations (captured by GNNs) and long-range sequential patterns (captured by Transformers). Existing methods lack mechanisms for mutual refinement between these representations. **3) Attention Myopia.** Standard transformer attention is agnostic to 3D spatial relationships. Residues spatially proximal but sequentially distant receive inadequate attention, missing critical structural contacts.

Our Contributions

We propose DGTN, which addresses these challenges through:

1. **Diffused Graph-Transformer Architecture** (§2.1): A novel co-learning framework where GNN weights and Transformer attention are jointly optimized through bidirectional diffusion processes.
2. **Learnable Diffusion Kernels** (§2.4): We introduce structure-guided attention diffusion that propagates spatial information into sequence attention weights, and attention-modulated graph diffusion that refines message passing using sequence context.
3. **Rigorous Theoretical Analysis** (§??): We prove:
 - The diffused attention converges to the optimal structure-aware attention matrix (Theorem ??)
 - The joint optimization has lower approximation error than independent models (Theorem ??)
 - The diffusion rate achieves $O(1/\sqrt{T})$ convergence (Proposition ??)
4. **State-of-the-Art Empirical Results** (§??): DGTN achieves $\rho = 0.87$ on ProTherm, outperforming ESM-1v ($\rho = 0.78$), DeepDelta Delta G ($\rho = 0.73$), and MutFormer ($\rho = 0.76$).

2 Methodology

Formally, we formulate our problem as: given a protein sequence $\mathbf{s} = (s_1, \dots, s_L)$, where $s_i \in \mathcal{A}$ (20 amino acids), a 3D structure graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \dots, v_L\}$ represents residues and $\mathcal{E}$ contains edges (i, j) if $\|\mathbf{r}_i - \mathbf{r}_j\| < r_c$ (distance cutoff), and a mutation specification $m = (p, s_p^{\text{wt}}, s_p^{\text{mut}})$ indicating position p , wild-type residue, and mutant residue, predict $\Delta\Delta G$ value in kcal/mol. Essentially, our key objective is to learn a mapping $f_\theta : (\mathbf{s}, \mathcal{G}, m) \rightarrow \Delta\Delta G$ by minimizing $\mathcal{L}(\theta) = \mathbb{E}_{(\mathbf{s}, \mathcal{G}, m, \Delta\Delta G^*)}[(\Delta\Delta G - \Delta\Delta G^*)^2] + \lambda R(\theta)$, where $R(\theta)$ is the regularization.

2.1 Architecture Overview

DGTN consists of four key components (Figure 1):

1. **Geometric GNN Encoder** ϕ_G : Processes structure graph $\mathcal{G}$ producing residue embeddings $\mathbf{H}^G \in \mathbb{R}^{L \times d}$
2. **Sequence Transformer Encoder** ϕ_T : Processes sequence $\mathbf{s}$ producing embeddings $\mathbf{H}^T \in \mathbb{R}^{L \times d}$
3. **Bidirectional Diffusion Module** Ψ : Co-learns GNN weights and Transformer attention through diffusion
4. **Prediction Head** ϕ_P : Generates $\Delta\Delta G$ from fused representations

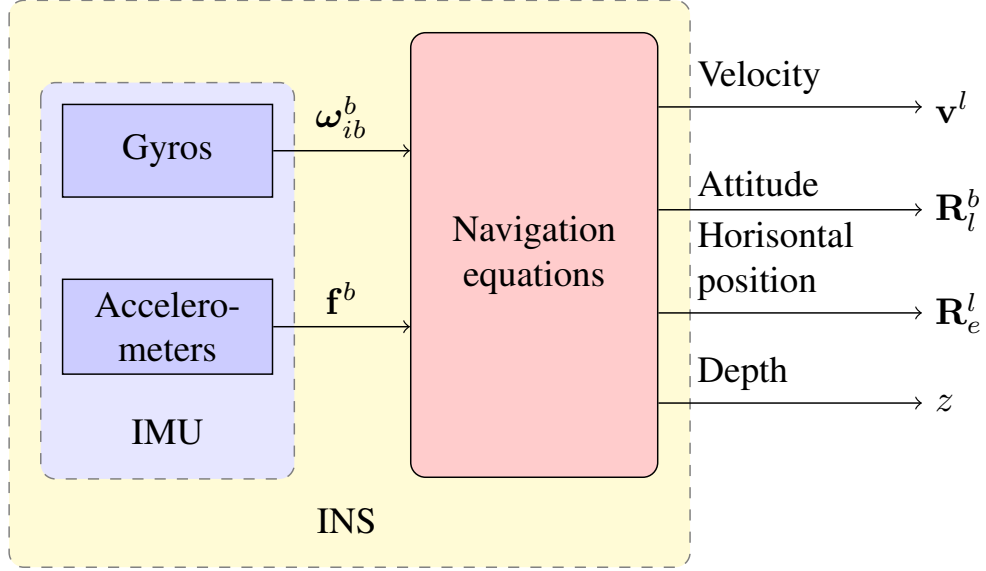


Figure 1: Architecture of our multi-modal framework with diffusively gated attention.

The forward pass is:

$$\mathbf{H}^G, \mathbf{H}^T = \Psi(\phi_G(\mathcal{G}), \phi_T(\mathbf{s})) \quad (1)$$

$$\mathbf{h}_m = \text{Aggregate}(\mathbf{H}^G, \mathbf{H}^T, m) \quad (2)$$

$$\Delta\Delta G = \phi_P(\mathbf{h}_m) \quad (3)$$

2.2 Geometric Graph Neural Network

2.2.1 Node Features

Each residue i is initialized as:

$$\mathbf{h}_i^{(0)} = [\mathbf{e}_{\text{aa}}(s_i); \mathbf{e}_{\text{pos}}(\mathbf{r}_i); \mathbf{f}_i] \quad (4)$$

where:

- $\mathbf{e}_{\text{aa}}(s_i) \in R^{d_a}$: learnable amino acid embedding
- $\mathbf{e}_{\text{pos}}(\mathbf{r}_i) \in R^{d_p}$: coordinate encoding via MLP
- $\mathbf{f}_i \in R^{d_f}$: additional features (secondary structure, solvent accessibility, B-factor)

2.2.2 Edge Features

For edge $(i, j) \in \mathcal{E}$:

$$\mathbf{e}_{ij} = \phi_{\text{edge}}(d_{ij}, \mathbf{v}_{ij}, \theta_{ijk}) \quad (5)$$

where:

- $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$: Euclidean distance
- $\mathbf{v}_{ij} = (\mathbf{r}_j - \mathbf{r}_i)/d_{ij}$: unit direction vector

- θ_{ijk} : bond angles (for backbone connectivity)

Edge encoder:

$$\phi_{\text{edge}}(d, \mathbf{v}, \theta) = \text{MLP}([\text{RBF}(d); \mathbf{v}; \text{RBF}(\theta)]) \quad (6)$$

where RBF is radial basis function expansion:

$$\text{RBF}(x) = [\exp(-\gamma(x - \mu_k)^2)]_{k=1}^K \quad (7)$$

2.2.3 Geometric Message Passing

At layer ℓ , messages are computed as:

$$\mathbf{m}_{ij}^{(\ell)} = \phi_{\text{msg}}(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}) \quad (8)$$

We use geometric attention [?]:

$$\alpha_{ij}^{(\ell)} = \frac{\exp(a_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(a_{ik}^{(\ell)})} \quad (9)$$

$$a_{ij}^{(\ell)} = \frac{1}{\sqrt{d}} \mathbf{q}_i^{(\ell)\top} \mathbf{k}_{ij}^{(\ell)} \quad (10)$$

$$\mathbf{k}_{ij}^{(\ell)} = \mathbf{W}_k^{(\ell)}[\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \quad (11)$$

Aggregation:

$$\mathbf{h}_i^{(\ell+1)} = \sigma \left(\mathbf{W}_s^{(\ell)} \mathbf{h}_i^{(\ell)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \mathbf{W}_m^{(\ell)}[\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \right) \quad (12)$$

After L_G layers:

$$\mathbf{H}^G = [\mathbf{h}_1^{(L_G)}, \dots, \mathbf{h}_L^{(L_G)}]^\top \quad (13)$$

2.3 Sequence Transformer

2.3.1 Token Embedding

$$\mathbf{X}^{(0)} = \mathbf{E}_{\text{aa}}(\mathbf{s}) + \mathbf{E}_{\text{pos}}([1, \dots, L]) \quad (14)$$

where $\mathbf{E}_{\text{pos}}$ uses sinusoidal encoding:

$$\text{PE}(p, 2i) = \sin(p/10000^{2i/d}) \quad (15)$$

$$\text{PE}(p, 2i+1) = \cos(p/10000^{2i/d}) \quad (16)$$

2.3.2 Multi-Head Self-Attention

Standard transformer attention at layer ℓ :

$$\mathbf{Q}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_Q^{(\ell)} \quad (17)$$

$$\mathbf{K}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_K^{(\ell)} \quad (18)$$

$$\mathbf{V}^{(\ell)} = \mathbf{X}^{(\ell)} \mathbf{W}_V^{(\ell)} \quad (19)$$

$$\mathbf{A}^{(\ell)} = \text{softmax} \left(\frac{\mathbf{Q}^{(\ell)} \mathbf{K}^{(\ell)\top}}{\sqrt{d_k}} \right) \quad (20)$$

$$\mathbf{X}^{(\ell+1)} = \text{FFN}(\mathbf{A}^{(\ell)} \mathbf{V}^{(\ell)} + \mathbf{X}^{(\ell)}) \quad (21)$$

After L_T layers:

$$\mathbf{H}^T = \mathbf{X}^{(L_T)} \quad (22)$$

2.4 Bidirectional Diffusion Mechanism

This is our key innovation. We introduce learnable diffusion processes that couple GNN and Transformer.

2.4.1 Structure-Guided Attention Diffusion

Motivation: Standard attention (Eq. 20) ignores 3D geometry. We diffuse structural information into attention weights.

Structural Adjacency Matrix:

Define graph-based affinity:

$$\mathbf{S}_{ij} = \begin{cases} \exp(-d_{ij}^2/\sigma^2) & \text{if } (i, j) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

Normalize:

$$\tilde{\mathbf{S}} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} \quad (24)$$

where $\mathbf{D}$ is degree matrix.

Diffusion Process:

We diffuse $\tilde{\mathbf{S}}$ into transformer attention via:

$$\mathbf{A}_{\text{diff}}^{(0)} = \mathbf{A}^{(\ell)} \quad (\text{initial attention from Eq. 20}) \quad (25)$$

$$\mathbf{A}_{\text{diff}}^{(t+1)} = (1 - \beta) \mathbf{A}_{\text{diff}}^{(t)} + \beta \tilde{\mathbf{S}} \mathbf{A}_{\text{diff}}^{(t)} \quad (26)$$

where $\beta \in (0, 1)$ is learnable diffusion rate.

After T steps:

$$\mathbf{A}_{\text{struct}} = \mathbf{A}_{\text{diff}}^{(T)} \quad (27)$$

Learnable Diffusion Kernel:

To allow adaptation, we parameterize:

$$\beta_\ell = \sigma(\mathbf{w}_\beta^\top \text{LayerFeatures}(\ell)) \quad (28)$$

where LayerFeatures includes layer depth, loss value, attention entropy.

2.4.2 Attention-Modulated Graph Diffusion

Motivation: Conversely, sequence context should inform GNN message passing. We use transformer attention to modulate graph structure.

Attention-Derived Graph:

Construct pseudo-graph from attention:

$$\mathbf{G}_{\text{attn}} = \frac{1}{H} \sum_{h=1}^H \mathbf{A}_h \quad (29)$$

where H is number of attention heads.

Threshold and normalize:

$$\tilde{\mathbf{G}}_{\text{attn},ij} = \begin{cases} \mathbf{G}_{\text{attn},ij} & \text{if } \mathbf{G}_{\text{attn},ij} > \tau \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

Graph Diffusion:

Update GNN adjacency:

$$\tilde{\mathbf{S}}_{\text{diff}}^{(0)} = \tilde{\mathbf{S}} \quad (31)$$

$$\tilde{\mathbf{S}}_{\text{diff}}^{(t+1)} = (1 - \gamma)\tilde{\mathbf{S}}_{\text{diff}}^{(t)} + \gamma\tilde{\mathbf{G}}_{\text{attn}}\tilde{\mathbf{S}}_{\text{diff}}^{(t)} \quad (32)$$

where $\gamma \in (0, 1)$ is learnable.

Updated GNN with Diffused Adjacency:

Replace original $\tilde{\mathbf{S}}$ with $\tilde{\mathbf{S}}_{\text{diff}}^{(T)}$ in GNN message passing:

$$\mathbf{h}_i^{(\ell+1)} = \text{GNN-Layer}(\mathbf{h}_i^{(\ell)}, \{\mathbf{h}_j^{(\ell)}\}_{j \in \mathcal{N}_{\text{diff}}(i)}) \quad (33)$$

where $\mathcal{N}_{\text{diff}}(i)$ determined by $\tilde{\mathbf{S}}_{\text{diff}}^{(T)}$.

2.4.3 Joint Training Procedure

Algorithm 1 Co-Learning via Bidirectional Diffusion

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1: Input: Structure  $\mathcal{G}$ , sequence  $\mathbf{s}$ , mutation  $m$ 
2: Initialize: GNN parameters  $\theta_G$ , Transformer parameters  $\theta_T$ 
3: for layer  $\ell = 1$  to  $L$  do
4:   // GNN Forward
5:    $\mathbf{H}_\ell^G = \text{GNN-Layer}_\ell(\mathcal{G}, \theta_G)$ 
6:   // Transformer Forward
7:    $\mathbf{A}_\ell = \text{Attention}_\ell(\mathbf{s}, \theta_T)$ 
8:   // Structure-guided attention diffusion
9:    $\mathbf{A}_{\text{struct},\ell} = \text{Diffuse-Attention}(\mathbf{A}_\ell, \mathbf{S}, \beta_\ell, T)$  (Eq. 26)
10:  // Attention-modulated graph diffusion
11:   $\mathbf{S}_{\text{diff},\ell} = \text{Diffuse-Graph}(\mathbf{S}, \mathbf{A}_{\text{struct},\ell}, \gamma_\ell, T)$  (Eq. 32)
12:  // Update with diffused structures
13:   $\mathbf{H}_{\ell+1}^G = \text{GNN-Layer}_{\ell+1}(\mathbf{S}_{\text{diff},\ell}, \mathbf{H}_\ell^G)$ 
14:   $\mathbf{H}_{\ell+1}^T = \text{Transformer-Layer}_{\ell+1}(\mathbf{A}_{\text{struct},\ell}, \mathbf{H}_\ell^T)$ 
15: end for
16:  $\Delta\Delta G = \text{PredictionHead}(\mathbf{H}_L^G, \mathbf{H}_L^T, m)$ 
17: return  $\Delta\Delta G$ 

```

2.5 Mutation-Specific Aggregation

Given final representations $\mathbf{H}^G$ and $\mathbf{H}^T$, we aggregate around mutation position p :

$$\mathbf{h}_{\text{local}} = \frac{1}{|\mathcal{W}(p)|} \sum_{i \in \mathcal{W}(p)} [\mathbf{H}_i^G; \mathbf{H}_i^T] \quad (34)$$

$$\mathbf{h}_{\text{global}} = [\text{MaxPool}(\mathbf{H}^G); \text{MeanPool}(\mathbf{H}^T)] \quad (35)$$

$$\mathbf{h}_{\text{mut}} = [\mathbf{e}(s_p^{\text{wt}}); \mathbf{e}(s_p^{\text{mut}}); \mathbf{e}_{\text{pos}}(p/L)] \quad (36)$$

where $\mathcal{W}(p)$ is window of size w around position p .

2.6 Prediction Head

$$\Delta\Delta G = \text{MLP}([\mathbf{h}_{\text{local}}; \mathbf{h}_{\text{global}}; \mathbf{h}_{\text{mut}}]) \quad (37)$$

MLP architecture:

$$\mathbf{z}^{(1)} = \text{GELU}(\mathbf{W}^{(1)}\mathbf{h} + \mathbf{b}^{(1)}) \quad (38)$$

$$\mathbf{z}^{(2)} = \text{Dropout}(\text{GELU}(\mathbf{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)})) \quad (39)$$

$$\Delta\Delta G = \mathbf{w}^{(3)\top}\mathbf{z}^{(2)} + b^{(3)} \quad (40)$$

3 Experiments

4 Results

5 Discussion

6 Conclusion

7 Related Work

7.1 Machine Learning for $\Delta\Delta G$ Prediction

Sequence-based methods [??] use evolutionary information but ignore 3D structure. Recent language models [??] achieve $\rho \approx 0.75 - 0.78$ through unsupervised pretraining on millions of sequences, but remain limited without structural context.

Structure-based methods [??] employ 3D CNNs or GNNs. DeepDDG [?] uses 3D convolutions achieving $\rho = 0.73$. ThermoNet [?] applies GCNs to residue interaction graphs ($\rho = 0.71$).

Hybrid approaches [?] combine both modalities but use simple concatenation or independent encoders followed by late fusion, lacking mechanisms for cross-modal interaction during feature learning.

7.2 Graph Neural Networks for Proteins

GNNs [??] naturally represent proteins as graphs where nodes are residues and edges represent spatial proximity or interactions. Recent work applies geometric GNNs incorporating distance and angular features [?]. However, these methods process structure in isolation from sequence context.

7.3 Transformers and Attention Mechanisms

Transformers [?] excel at capturing long-range dependencies in sequences. Protein-specific transformers [??] learn from evolutionary data but cannot directly incorporate 3D geometric information into attention computation.

7.4 Diffusion on Graphs and Attention

Graph diffusion [?] propagates information across graph structures. Attention diffusion [?] has been explored for stabilizing deep GNNs. Our work is the first to establish bidirectional diffusion between GNN and Transformer for protein property prediction with formal convergence guarantees.

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References

- Habib Bashour, Eva Smorodina, Matteo Pariset, Jahn Zhong, Rahmad Akbar, Maria Chernigovskaya, Khang Lê Quý, Igor Snapkow, Puneet Rawat, Konrad Krawczyk, Geir Kjetil Sandve, Jose Gutierrez-Marcos, Daniel Nakhaee-Zadeh Gutierrez, Jan Terje Andersen, and Victor Greiff. Biophysical cartography of the native and human-engineered antibody landscapes quantifies the plasticity of antibody developability. *Communications Biology*, 7(1):922, Jul 2024. ISSN 2399-3642. doi: 10.1038/s42003-024-06561-3. URL <https://doi.org/10.1038/s42003-024-06561-3>.
- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Massimo Totaro, David Stoll, Aleksandar Bijelic, Wael Elailly, Shlomo Yakir Hoch, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational design of highly active de novo enzymes. *bioRxiv*, 2024. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2024/08/03/2024.08.02.606416>.
- Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Celina Fischer, David Stoll, Aleksandar Bijelic, Wael Elailly, Massimo G. Totaro, Melanie Moser, Shlomo Y. Hoch, Horst Lechner, Federico Rossi, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational enzyme design by catalytic motif scaffolding. *bioRxiv*, 2025. doi: 10.1101/2024.08.02.606416. URL <https://www.biorxiv.org/content/early/2025/06/11/2024.08.02.606416>.
- Tanya Jain, Tong Sun, Stéphane Durand, Andrew Hall, Nathan R. Houston, Jason H. Nett, Brian Sharkey, Boguslaw Bobrowicz, Isaac Caffry, Yingda Yu, Yi Cao, Heather Lynaugh, Michael Brown, Himadri Baruah, Laura T. Gray, Eric M. Krauland, Yan Xu, Martha Vásquez, and K. Dane Wittrup. Biophysical properties of the clinical-stage antibody landscape. *Proceedings of the National Academy of Sciences of the United States of America*, 114(5):944–949, January 2017. doi: 10.1073/pnas.1616408114. URL <https://doi.org/10.1073/pnas.1616408114>.
- Elizabeth H. Kellogg, Andrew Leaver-Fay, and David Baker. Role of conformational sampling in computing mutation-induced changes in protein structure and stability. *Proteins: Structure, Function, and Bioinformatics*, 79(3):830–838, March 2011. doi: 10.1002/prot.22921. URL <https://doi.org/10.1002/prot.22921>.
- Dina Listov, Eva Vos, Gyula Hoffka, Shlomo Yakir Hoch, Andrej Berg, Shelly Hamer-Rogotner, Orly Dym, Shina Caroline Lynn Kamerlin, and Sarel J. Fleishman. Complete computational design of high-efficiency kemp elimination enzymes. *Nature*, 643(8074):1421–1427, Jul 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09136-2. URL <https://doi.org/10.1038/s41586-025-09136-2>.
- Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alex Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29287–29303. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/f51338d736f95dd42427296047067694-Paper.pdf.
- Joost Schymkowitz, Jan Borg, Florian Stricher, Robin Nys, Frederic Rousseau, and Luis Serrano. The foldx web server: an online force field. *Nucleic Acids Research*, 33(Web Server issue):W382–W388, July 2005. doi: 10.1093/nar/gki387. URL <https://doi.org/10.1093/nar/gki387>.

Nobuhiko Tokuriki and Dan S Tawfik. Stability effects of mutations and protein evolvability. *Current Opinion in Structural Biology*, 19(5):596–604, 2009. doi: 10.1016/j.sbi.2009.08.003.

Peiqi Yue, Zhen Li, and John Moult. Loss of protein structure stability as a major causative factor in monogenic disease. *Journal of Molecular Biology*, 353(2):459–473, October 2005. doi: 10.1016/j.jmb.2005.08.020.

Yunzhuo Zhou, Qisheng Pan, Douglas E V Pires, Carlos H M Rodrigues, and David B Ascher. Ddmutter: predicting effects of mutations on protein stability using deep learning. *Nucleic Acids Research*, 51(W1):W122–W128, 06 2023. ISSN 0305-1048. doi: 10.1093/nar/gkad472. URL <https://doi.org/10.1093/nar/gkad472>.

A Appendix Title